

True-view videoconferencing system through 3-D impression of telepresence

L-Q Xu, A Loffler, P J Sheppard, and D Machin

It can be argued that an ideal teleconferencing system would offer a sense of presence comparable to that afforded by real face-to-face meetings. In contrast, traditional desktop videoconferencing loses essential socially important phenomena such as true perspective views, live facial expressions and tacit body language. This paper reports a prototype concept demonstrator under development at BT Laboratories that addresses these issues in order to make the use of videoconferencing a normal social activity rather than a means of merely getting messages across the channel. The idea is to place participants from different remote sites in a coherent virtual environment naturally extended from the real world. The participants have correct views of each other and the illusion of close proximity. The rationale is to build on the demonstrator and enhance it with increasingly realistic functional components as the relevant technical solutions become available. Results are presented of a stereo-based vision system which can synthesise novel (or virtual camera) perspective views of the scene from two fixed camera views. The technique is currently capable of providing 'look-around' views and a certain degree of navigation capability around the scene. Research issues are discussed together with the practical difficulties that remain to be solved in implementing a true-view, telepresence, videoconferencing system.

1. Introduction

Our goal for telepresence videoconferencing is to create a meeting space where virtual and real worlds are seamless. The participants would view each other from the proper perspective, with realistic eye contact, and the individual viewpoint would change appropriately as people move about within the common meeting environment. Traditional videoconferencing does not support important social and psychological elements of meetings, such as the spatial relationship between people in a meeting room and directed body language.

One method that has been successfully applied to create realistic eye contact has been the mechanical arrangement of a camera placed behind a semi-transparent screen [1]. Here the camera can be positioned behind the eyes of a person's image. Other systems use separate monitors for each participant. In Fields [2], the monitors are small and placed on a desktop each with its own camera so that the gaze direction is almost correct. In Travis et al [3], a room-based environment is created using larger monitors with cameras placed at strategic locations around the room. An alternative approach [4–7] uses 3-D virtual human models that look like the participants in the meeting. The facial expressions and arm movements are tracked using physical tracking devices or computer vision techniques, these movements being replicated by the virtual humans. This technique may be useful for large meetings where achieving

correct inter-person angles is difficult using video or for enhancing audio conferences.

While the approaches described above attempt to re-introduce into videoconferencing some of the social and interaction cues that are present in physically collocated meetings, they also introduce barriers to the illusion of presence between the participants, e.g. non-life-size images, low degree of realism of visual rendering, discontinuous transition between local and remote environments, eye-to-eye contact limited by physical set-up, body language lost by having only head and shoulder shots, fixed 2-D views. A concept demonstrator for telepresence videoconferencing is presented in which a user feels immersed in a common environment with remote participants. This is achieved by correct registration of the virtual with the real environment and by displaying true gaze directions, using image processing techniques to create intermediate camera views.

From a technical point of view, the main challenge is creating novel views of the remote, dynamically changing scene with minimal visual artefacts. These intermediate views are synthesised from those obtained by the fixed cameras. This enables the observer to experience a 'look-around' view of the scene (the illusion of motion parallax) when he moves his head. Importantly it will also make it possible for the observer to establish direct eye contact with

remote partners, and gain awareness of gaze contacts between other participants when more than two locations are involved in such a conference.

The techniques used for composing the true-view synthesis system belong to a class of image-based rendering paradigms [8]. Other popular methods are based on 3-D model reconstruction from the reference images allowing rendering of the final model from any desired viewpoint [9, 10]. Recent research has shown that, under certain conditions, a discrete set of images from different viewpoints can implicitly represent scene appearance for a continuous range of viewpoints [11]. There are a variety of methods being proposed for the task of view synthesis in a general sense using two or three cameras [12–15]. The complexity of the algorithms and the practical issues of this problem domain (aiming for a semi-immersive conferencing system) demands special attention in choosing a specific paradigm. The method adopted in this paper uses an object-based hierarchical disparity estimation and scene-based interpolation strategy similar to that used in Ohm and Izquierdo [16]. The advantage of image-based approaches is a naturalness that is hard to achieve with graphic objects, additionally the rendering time is not proportional to the complexity of a model, but only the number of pixels, an advantage for real-time applications. It is noted however that for difficult scenarios (section 3) a solely image-based approach has its own limitations that are demonstrated in this paper and will be addressed in future work.

2. True-view videoconferencing system

The true-view videoconferencing system is intended to enhance the naturalness of videoconferencing by employing state-of-the-art technologies in computer vision and video processing. It features several important advances.

- Integrated multiparty (multi-site) conferencing — a meeting held simultaneously between more than two sites around the world will show the participants seated around a common virtual meeting table, a through-the-screen extension of their own real table. This is in contrast to current conferencing technologies that display each participant on a separate screen or in a separate window,
- Illusion of telepresence — participants are viewed life-size (on a large semi-immersive display) within the integrated virtual meeting space, and correct gaze directions are reproduced throughout, hence ‘true-view’. This means that the participants can establish true personal eye contact towards each other and successfully use directed body language.
- 3-D viewing effects — participants can move their head position and look around the scene, viewing different parts of the virtual meeting room (including different views of the participants in the final system).

Motion parallax will give the users a feeling of depth. Ultimately this can be extended to a true auto-stereoscopic 3-D display.

The virtual environment could, for example, mimic a multinational corporate boardroom — the partners would have the feeling of talking to each other in close contact around the conference table. The system brings additional, but essential, social elements into a conference environment in the sense that the participant at one site can tell the keenness or boredom of others from their body language and interactions — as with a face-to-face conference (see Fig 1).

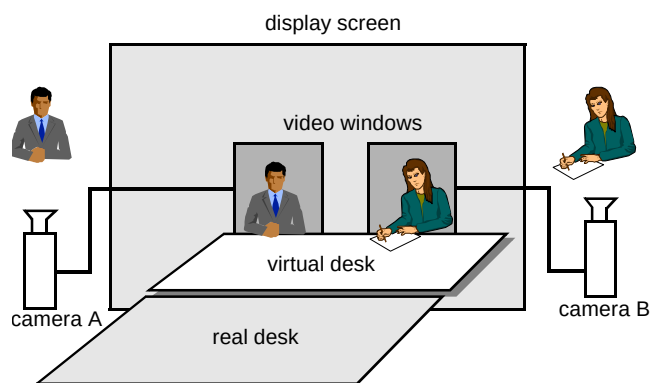


Fig 1 Demonstration of natural conferencing system. The user to appear in front of the real desk would have the feeling of collocation with remote partners.

To demonstrate the concepts and to investigate the usability issues of a true-view conferencing system two demonstrators were created:

- desk-sized version (Fig 2) — featuring a state-of-the-art, 42-inch, flat plasma display with no head position tracking,
- room-sized version (Fig 3) — featuring a large back-projected display screen and a real-time head tracker.

In both systems, a participant sat at a desk that extended seamlessly from the real to the virtual scene that was displayed on a screen at the end of the desk (Fig 1). Video windows with the remote participants segmented from their background scene were placed around the virtual part of the desk. This enabled creation of the illusion that everyone was in the same room and the ‘location’ of the meeting could be changed by selecting different background images — e.g. one moment we could be in Sydney Harbour (Fig 2) the next on the moon. At present only one user’s site has a special display screen from which Figs 2 and 3 are taken respectively; the remote users viewed using small monitors and did not have the full feeling of presence.



Fig 2 View of the system using the 42-inch plasma display mounted on a black desk. All but the leftmost coffee cup and the keyboard on display are virtual objects.



Fig 3 View of the system using a back-projected screen. A real desk is placed in front of the screen and user sits in front of it to converse with the two remote participants.

Traditional chroma-key techniques initially appeared to be the most attractive for combining the video of the participants with a backdrop, though shadows of the hands and arms on the desk caused a problem (explained below), and, in practice, a combination of chroma-key and background subtraction proved to be the most successful. In this technique, the system is initially given the views of an 'empty' scene consisting of a green backdrop and a desk of a different colour. The participant is then placed in the scene along with their chair and any other items (e.g. coffee cups). An absolute subtraction operation on the current image and the stored 'empty' image is performed resulting in a pixel by pixel difference mask. This mask can be used as a filter to remove any background information, leaving only foreground objects, i.e. the subject complete with chair, items and any shadows cast.

Shadows cast by the subject (predominantly those cast by the arms on to the desk) are not filtered out by the current system (this is an inherent effect of the differencing strategy used). One way of addressing the problem is to ensure the colour of the participants' desk is exactly matched to that in

the virtual world. It was found that black was an ideal choice (it also had the advantage of removing the halo effect induced by chroma-keying against a green background).

The large-screen demonstrator shown in Fig 3 also incorporated an image-based head tracker that was used to adapt the views to the user of the 3-D virtual world on display — the virtual table and the plant in this case. When the user moves his head he can experience the changes-of-view angles of the virtual world. It is, however, noted that the views of the two remote participants are still fixed dependent on the view angles of two remote video cameras respectively¹.

Use of these systems has highlighted several important issues.

- Camera position

In a system where all participants have the same type of large screen, it would be necessary to have cameras positioned behind the screen (or 'invisibly' in front of the screen) in order to achieve correct eye contact. A mechanical arrangement of cameras creates a number of difficulties including limiting the flexibility of participant location, scaling to a larger number of participants and recreating fast, dynamic view shifts (look-around effects). The work on interpolated view camera systems described in subsequent sections is therefore essential.

- Viewing position

To provide a convincing, and non-distracting, look-around effect the system must track the user's head position and redisplay the scene with minimal delay. In this implementation the tracking was fast but the re-display response speed was limited by the image-rendering hardware. The ability to maintain correct motion parallax (e.g. the edges of the real and virtual desks are seen to converge towards the same point) did, however, have a powerful influence on the feeling of through-the-screen presence.

- Size of screen

To create the effect of immersion in the virtual environment a large wrap-around screen would be ideal. For the system with the larger display screen (Fig 3) the feeling of presence is stronger than with the 42-inch plasma screen.

- Size of video windows

It is essential that the sizes of the people in the image are correct for their positions in the world. If the desk looks 1 metre long but the video windows suggest that

¹ In fact, to acquire a realistic synthesised view of the remote participants, that responds to the user's head movement, is a central issue to be discussed in the rest of the paper.

the people are 3 metres away then the feeling of presence disappears. Interestingly, by reducing the size of one video window slightly, the dominance of that person seemed to be reduced.

3. Demonstration of novel view synthesis

As explained previously the ability to synthesise novel views is essential to create the illusion of telepresence. In this study the novel (virtual) camera views are defined as being along the interocular line between the two fixed cameras actually capturing the images of the scene. A dynamic scene from the videoconferencing set up, picturing one human subject with an appropriate background, is used. The subjects can change their sitting positions and make various facial, hand and body gestures as one normally does during a face-to-face conversation. This section demonstrates the performance of the system while leaving the detailed descriptions of these techniques to the next section.

In particular, with still images taken from two cameras positioned along a line (with a reasonably small convergence angle), the synthesised views of the scene at different viewpoints are shown along the baseline between the two cameras. The established sequence thus presents a limited panoramic (shifted + rotated) view of the 3-D scene. A fixed (albeit arbitrary) view of the scene is synthesised for the duration of the recording. Image sequences ranging from simple head-and-shoulder shots through to more difficult but realistic scenarios are discussed below.

3.1 Head-and-shoulder scene

The head and shoulder scene is a popular test scene for a telepresence videoconferencing system. Because the person's head occupies a relatively large area of the frame, it is easy to verify if, in a synthesised new viewpoint, the true eye contacts as well as details of facial expressions are realistic, hence certifying the validity of the algorithm. The two still images (left and right view) shown in Fig 4 are from a European DISTIMA stereoscopic image pair. The images were shot with the two cameras separated by a baseline of $b = 45$ cm. Each camera has a focal length of $f = 18$ mm, and the convergence distance between the subject and cameras baseline was about $d = 2.25$ m. The original images obtained have an image size 288 by 720 pixels².

From these two images only, without the need of an explicit 3-D model or 3-D scene reconstruction, this system can reproduce the scene from any viewpoint on the interocular line between the two cameras with minimum perceivable visual artefacts. Figure 5 is the interpolated middle view between the two original cameras, and shows

the correct head orientation, facial shape, pose and desired eye-to-eye contact besides the true colouring effects, despite the fact that only 2-D image operations have been performed. Note that small alias errors have occurred at the bottom of the picture and that a very small patch of registration error appears in a few frames on the right hand side of the neck. These do not, however, cause any annoyance in viewing the temporally smoothed sequence.

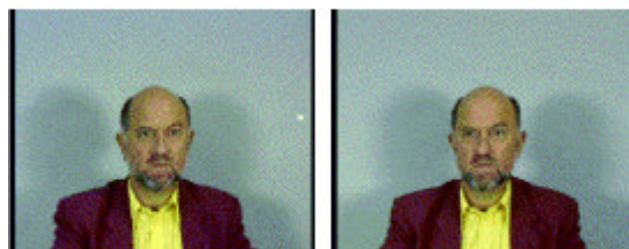


Fig 4 Example of DISTIMA stereoscopic image pair, left and right view respectively.



Fig 5 The synthesised virtual camera view image at the middle viewpoint.

The success of this system on these test images is helped by the following. Firstly, the 18 mm focal length of the lens used is relatively long, giving low image distortion. Secondly, the head-and-shoulder scene has a limited occlusion (i.e. most of the scene is visible from both viewpoints), though the subject's left ear is barely seen in the left view image and his right ear is largely occluded in the right view image. Thirdly, the distance between the subject and the cameras is long relative to the separation of the cameras, giving rise to smaller depth changes over the subject's convex surface.

3.2 Beyond the head-and-shoulder scene

Still images are useful for illustrating the functions of the virtual camera system, though we are more interested in the scenarios beyond the extent of head and shoulders in which a subject is allowed to make free body and hands gestures. For this purpose, a test bed was set up using three cameras arranged in parallel positions on a straight line to record various image sequences in a typical videoconferencing environment.

Figure 6 depicts a typical set up where C_0 and C_1 are the optical centres of two cameras taking images of the scene and C_v is the optical centre of a third camera pointing to the

² Note that the camera set-up information is for illustrative purposes only, as in the rest of the paper no camera calibration is carried out for the results presented.

same direction. I_0 , I_1 and I_v represent the corresponding image planes. Only the left and right view camera image sequences are used in the experiments. The actual image sequences captured by the third middle view camera are used for comparison purposes afterwards. The convergence distance between a subject and the plane of the two cameras is in the range of $d = 120$ cm and 160 cm. Compare this set-up, especially the short focal length and small object distance, with that of the head-and-shoulder scene above. In this case the task of synthesising a realistic view becomes very difficult; the situation is exacerbated when the baseline becomes larger and the scene contains joint objects.

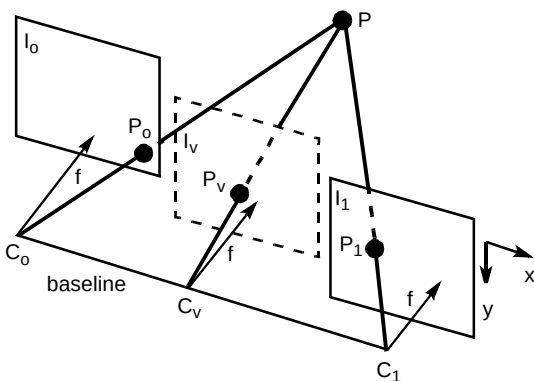


Fig 6 The parallel camera set-up for recording test images sequences. C_0C_1 is the interocular line. C_0 and C_1 represent, respectively, the optical centres of the two cameras capturing images of the scene; C_v is the optical centre of a third parallel camera.

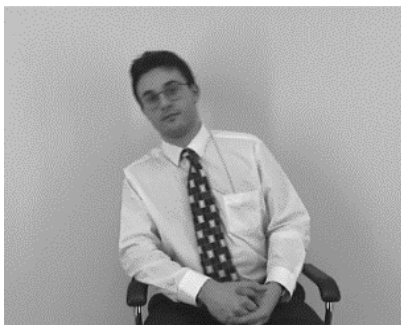


Fig 7 The left and right camera view (frame 45) of a video sequence Alex24.



Fig 8a The real middle camera view at Frame 45 of sequence Alex24. Note the turned head orientation in comparison with that in Fig 7.

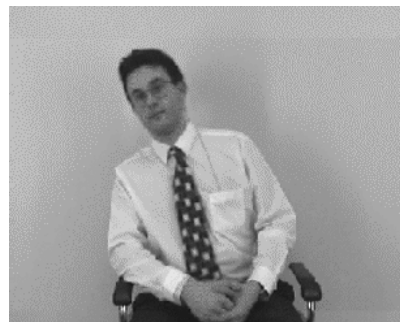


Fig 8b The corresponding synthesised virtual camera view.

3.3 Cameras with a 24 cm separation

A sequence (Alex24) of 60 frames was shot with the two cameras at a distance of 24 cm apart and a third, halfway between the two. Typical images (frame 45) are shown in Fig 7 for the left and right camera views and in Fig 8a for the middle camera view.

For comparison, we present in Fig 8b the synthesised virtual middle camera view for the frame in question. Again, its realism can be demonstrated when comparing the two images (note only the trivial artefact on the subject's right hand shirt cuff).

It is straightforward for the current system to compose a dynamic temporal sequence whether the virtual camera keeps the same or continuously changes its viewing position of the scene through the period of recording time.

3.4 Cameras with a wide 50 cm separation

Further investigation was undertaken for the virtual camera system with image sequences taken by cameras with a wide 50 cm baseline. This is an important step towards a telepresence videoconferencing system working with an unrestricted camera set-up and (semi-) immersive display. The current results and the challenging issues that result are reported below. Figures 9 (left), 9 (right) and 10a show the images of the left, right and middle camera views for frame 25 of a recorded video stream Phil50. In Fig 10b the synthesised middle view is presented for the same frame. It can be observed, however, that visual artefacts are apparent

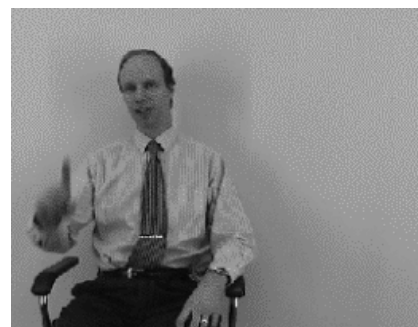


Figure 9. The left and right camera view (frame 25) of a video sequence Phil50. Note especially that large disparities exist, as the left view image is near the right edge of the frame while the right view image is near the left edge of the frame. Note also the occlusions caused by the raised arm and hand in each of the two views — an infamous problem in view synthesis of a small narrow object in front of a large background object.

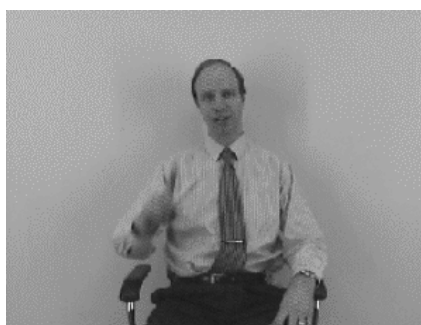


Fig 10a The true middle view camera shot of the scene for frame 25 of sequence Phil50.



Fig 10b The system-synthesised virtual middle camera view. Apparent visual artefacts found exist in the occluded body areas.

with this case, especially in the body areas that can only be seen from one of the two original viewpoints. The areas affected include the raised right hand, the right arm and the patch between the left hand and body, in addition to the fact that the facial details are blurred.

Apart from the complexities of the dynamic scene in which a speaker quickly moves his hands and body, partial failures of the system are due mainly to the following two facts. Firstly, difficulties are caused by the short focal length requiring a close distance between the subject and the cameras, discussed in section 3.2 (though this small camera-object distance is true for a semi-immersive display). Secondly, the two-stage framework (discussed in section 4) has its own limitations that are yet to be improved to deal with the issues exposed here.

4. The stereo-based view synthesis system

The system used for the synthesis of novel views side-steps the process of explicit 3-D scene reconstruction and back-projection, arguably avoiding the difficulties that involves. It does so within a two-step, object-based framework. The first step (section 4.1) is to compute the dense disparity fields between the segmented objects appearing in the two (basis) images taken from different viewpoints (with knowledge of camera parameters). The disparity field provides the essential information of correspondence between the left

and right images, or it tells which part (pixel) of the left view image corresponds to which part (pixel) of the right view image. The second step (section 4.2) is to interpolate on a pixel by pixel basis the new image as though it has come from a given new viewpoint, based on the recovered disparity information. This framework has been adopted by the European PANORAMA project [16] and is quite appealing in the context of videoconferencing. However, it is open to some specific problems described above, and in particular the occlusion issues which will be explained in section 4.3.

4.1 Disparity estimation

Disparity estimation starts with a segmentation of the foreground subject from the uniform background by identifying and masking those interest points in both the left and right images.

A global block matching (left-to-right), operated on down-sampled images, is then carried out to associate the points in the masked right image with the identified interest points in the left image using a full search method. For each matched point found, an opposite (right-to-left) matching procedure is started to check the validity of the matching or the global disparity computed. Figure 11 gives an example of the masked left and right images with the identified global interest points and matched disparity field.

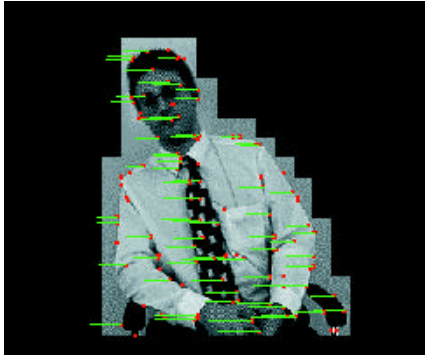


Fig 11 The global interest feature points and corresponding disparity fields.

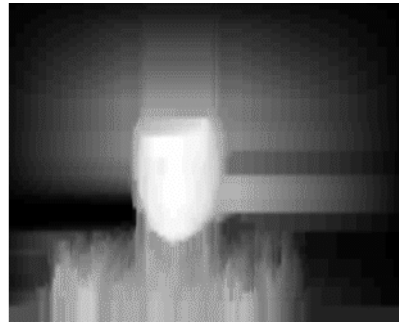
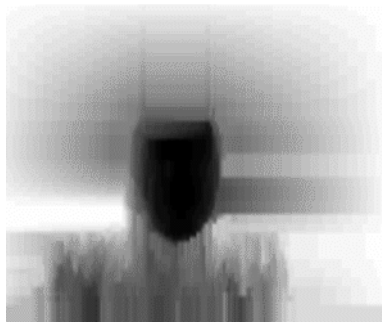


Fig 12 The left-to-right and right-to-left dense disparity fields estimated for the image pair in Fig 4.

Based on the global disparity fields above, a local block matching (left-to-right) is attempted for pixels on every 4th scan-line and 4th column if they lie inside the masked object. The search range in this case is decided by the neighbouring global block disparities, and the local block disparities just computed as well as temporal disparity prediction in the previous frame. Similarly, a right-to-left matching is found to cross-check the validity of this result.

The local disparity fields thus obtained are median filtered before being linearly interpolated column-wise and row-wise to form the desired dense disparity fields. Figure 12 demonstrates an example of the final left-to-right and right-to-left dense disparity fields; the disparity fields are smoothed from the object edges to the rest of the background. An error occurs at the left ear of the subject due to occlusion.

4.2 Interpolation strategy

Both the left-to-right and right-to-left disparity fields, denoted hereafter as DL and DR respectively, are used for interpolation for a new viewpoint. Although most information is largely redundant, the two disparity fields complement each other, showing different features in the occluded scene, as will be explained in section 4.3.

Assuming a left camera image at position $s = 0$ and a right camera image at position $s = 1$, the intermediate view with $0 < s < 1$ can be synthesised using either or both images

based on the proportion of the corresponding disparity fields. A usual strategy is that when $0 \leq s < 0.5$, the left image and DL is used, while the right image and DR is used in the case of $0.5 \leq s < 1$. In our system, however, the object mask available through previous segmentation is analysed to define, for different areas in the new image, which of the original two images is most appropriate.

4.3 The occlusion issue

Occlusion poses the main obstacle for the efforts described above. The nature of these problems can be appreciated from the sketches shown below. This can partly explain why occasional failures occur for the system in certain situations.

Figure 13 (left) depicts the left-to-right disparity path and the areas in the scene where left occlusions occur. In the figure, L and R denote respectively the left and right view camera. The scene consists of three objects — A lies in the background, B is in front of A , and C is in front of both A and B . d_a , d_b and d_c indicate the estimated depth when the scene can be visible from both cameras. For the estimation of left-to-right disparity, the left camera firstly scans across the scene from left to right. This is followed by the right camera acting in the same way; a check is then made to see if the right camera can see the same part of the scene that is already visible by the left one — a possible left-to-right match can then be made at a certain depth.

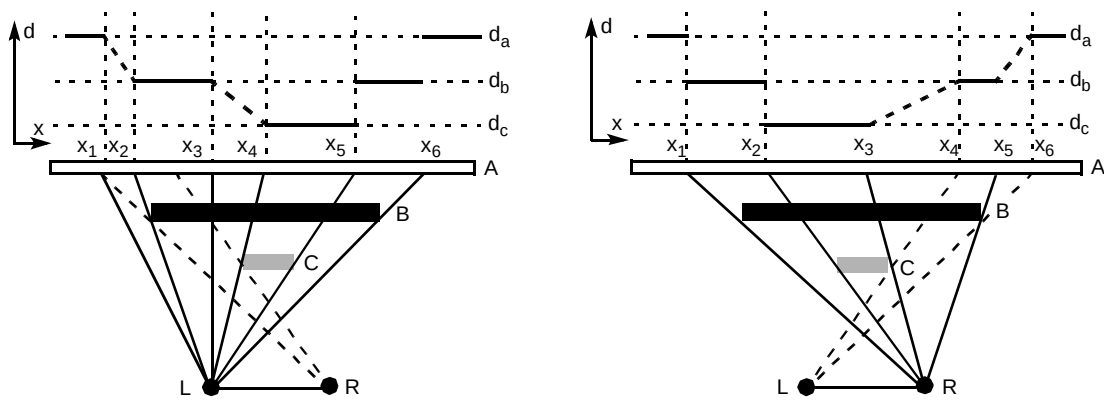


Fig 13 The left-to-right disparity path and the left occlusion (left); the right-to-left disparity path and the right occlusion (right).

In Fig 13, for the segment to the left of x_1 , both the left and right camera can see object A; a possible match is obtained at the depth d_a . When the left camera continues scanning to the right of x_1 within the segment of x_1x_2 , the right camera cannot see the object A; a left occlusion thus occurs. Similar observation applies to the segment x_3x_4 where the left camera can see object B while the right camera cannot, causing another left occlusion.

Similarly, in Fig 13 (right), for the estimation of right-to-left disparity, the right camera first scans across the scene from left to right; it is followed by the left camera acting in the same way. An attempt is then made to see if the left camera can see the same part of the scene, hence obtaining a possible right-to-left match. Right occlusions occur for the segment x_3x_4 and x_5x_6 where the right camera can see objects B and A, respectively, while the left camera is blocked.

Notice that the left-to-right and right-to-left disparity path shows different features of occlusion and jump. This information can be used in interpolating a new view to generate an appropriate interpolated image mapping, partially tackling the occlusion issue; though in practice any arbitrary situation can happen in the scene. One pathological case is exemplified in Fig 14 where the left-to-right disparity path is depicted. A small narrow object C lies in front of object A. Left occlusion occurs in the segment

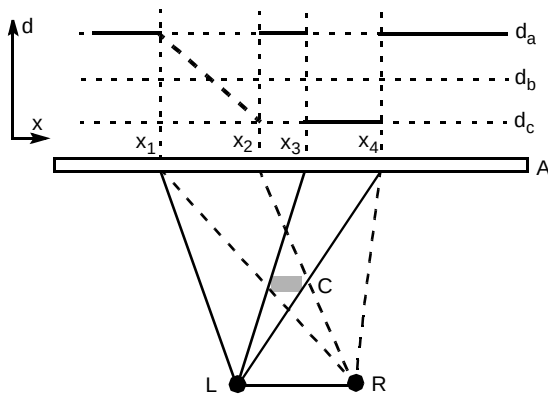


Fig 14 A demonstration of the violation of ordering constraints.

x_2x_3 similar to the situation described in Fig 9, but the right camera can see the segment x_2x_3 in object A around the object C. Thus, a possible match can also be made in this part of the image. This is a peculiar situation when the left camera can see the part in object A from left of object C, while the right camera sees it from the right of object C, thus violating the ordering constraints which is vital for both stereo vision and novel view synthesis. This is actually the single main reason for the visual artefacts of synthesised view in Fig 10b.

5. Discussion and conclusions

The goal of our true-view videoconferencing research has been described together with two demonstration systems built to illustrate the concepts and to investigate usability issues. True-view videoconferencing has the potential to greatly enhance the degree of presence experienced by participants in a multipoint videoconference. In addition, the enhanced ability to use social cues that are possible in a physically collocated meeting, but not possible in traditional videoconferencing, will enable a greater range of meetings to be successfully held remotely.

A system for re-creating the views required for such a true-view videoconferencing system has also been described. For simple dynamic conferencing scenes, the current system works well, but for more realistic scenarios the system partially failed. Nevertheless, there still exists scope for further improvement of the proposed algorithms [17].

Alternative strategies may give better results. An interesting development is the so-called 3-D video sprites that use trinocular stereo matching to provide a morphing channel in addition to the three synchronous video streams [18]. The matching is edge-based and relies on the extraction of suitable epipolar geometry. The resulting edge correspondences are then interpolated and used in an efficient morphing algorithm that operates one scan-line at a time to perform image synthesis. This approach will be further examined.

Recent rapid advancement in techniques, tools and hardware may mean that real-time 3-D scene rendering is no longer a major issue; this partly reflects the resurgent interest in 3-D scene reconstruction.

One idea to be investigated is to use two stereo camera pairs looking at the 3-D scene. Each pair contains two closely spaced cameras (a small baseline) to secure a better disparity estimation; the disparity fields obtained are then used to reconstruct the scene using an appropriate strategy. It remains to be seen how viable it is to fuse the two individual models to form a coherent model, especially for the articulated objects of interest here.

The concept of active stereo has not, as yet, been thoroughly investigated. Here, patterns of infra-red are focused on to the scene. Infra-red cameras fitted with suitable gratings are used to detect the projected patterns. Moiré fringes are recorded by the camera and techniques to measure object distance then become possible. This could be less problematic than existing disparity estimation techniques.

As yet, research has focused on the off-line processing of video sequences, each frame taking several seconds to process, even with optimised algorithms. For a practical system, each frame must be created in less than 40 ms. This may be difficult using a software-only solution and the development of a hardware accelerator is almost inevitable. It is also planned to investigate 3-D display technologies [19] for a real 3-D presence.

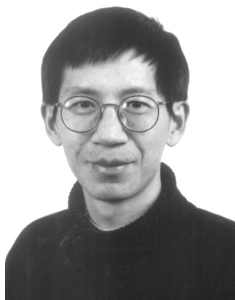
This work represents a new direction. An important next step is to evaluate users' reactions to video images created using the novel techniques described here. This evaluation of the system will build upon recent work [20] which has demonstrated the benefits of a multiparty videoconferencing system in which participants see each other in spatially consistent directions (so-called isotropic teleconferencing). This research suggests that persuasive force is significantly stronger under isotropic conditions than under non-isotropic conditions.

In conclusion, a concept demonstrator has been created to investigate the aspects of immersion and display technology that will be required in a commercial system. This demonstrator has highlighted the importance of correct gaze direction and the blending of the real and virtual environments to create a shared social space.

The ability to create intermediate views from two image sequences taken using widely separated cameras has been demonstrated. The algorithms are adequate for head-and-shoulder images, but do not as yet perform reliably when the hands occlude the face.

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Li-Qun Xu holds a BSc and an MSc in Electrical Engineering, and a PhD in Information Engineering from Southeast University, China, in 1988. He then worked for two years in the Institute of Acoustics, the Chinese Academy of Sciences, in Beijing. He joined the University of East Anglia in early 1990, supported by a Royal Society/BT Fellowship. Since then he has worked as Research Fellow at the University of Sussex and King's College, London, and Lecturer at the University of Abertay, Dundee. He joined BT Laboratories (BTL) in October 1996, initially working in the Intelligent Systems unit, and a year later in the Advanced Perception unit. He currently leads the Telepresence Campaign team at BTL responsible for researching into the key techniques and algorithms in 3-D computer vision and image-based rendering paradigm, with a view to developing an advanced true-view videoconferencing system.



After graduating with a degree in 1996 in Cognitive Science from Sheffield University, Alex Loffler joined BT Laboratories. He started work on real-time image processing, audio manipulation and video streaming (over ISDN-6 and 30). This culminated in the designing and building of the immersive 'True-View' conferencing system. Other projects of note include designing, building and administration of the BT intranet site for the BT Shareprice. He is currently working on a PhD in the field of wearable computing focusing on distributed co-operative agent-based infrastructures for building context-aware devices capable of intelligently altering their behaviours to suit a user's requirements



Phil Sheppard graduated from Birmingham University in 1986 with a BSc (Hons) in Computer Science and Electronic Engineering. He joined BT Laboratories on graduation where he has worked on ISDN transmission over copper, speech coding and recognition, network emulation, algorithm development for customer opinion modelling and non-intrusive measurement of voice networks for which he is an ITU-T Rapporteur. He currently leads a group working on a diverse set of projects to provide natural communications systems — including research into non-intrusive measurement techniques, conferencing systems (including VR technology and video image processing) and development of BT's 3-D Talking Head. He gained an MSc in Telecommunications Engineering from the University of London in February 1997.



Dave Machin gained a degree from the University of Bradford in 1975, joining BT Laboratories to work on mathematical modelling of transistors for integrated circuits. After a number of years in this area he spent five years on a collaborative European project for automatic design validation of integrated circuits using electron beams.

In 1990, he joined a team working on subjective testing of telephony services and developed a tool using digital signal processors to emulate international telephony networks. He has recently moved into the area of machine vision for teleconferencing.